Lot-Sizing Algorithms

ChronoForge Pulse | How EOQ, Wagner-Whitin and Silver-Meal consume the EDA features

This document explains exactly how the three lot-sizing algorithms used by ChronoForge Pulse turn the cleaned telecom dataset into procurement decisions. For each algorithm: the formula, which data feature feeds which input, a worked example using real Grameenphone numbers from **telecom_clean.csv**, and the conditions under which the algorithm is the right choice.

From EDA output to order plan

```
telecom_clean.csv  ->  ARIMA / LSTM / XGBoost  ->  monthly demand vector  ->  lot-si
zing  ->  Q* + timing
   (P2 output)            (P3, ensemble)              (per operator, per SKU)      (
3 algos)        (procurement)
```

Feature -> algorithm input (overview)

Our feature	Role	How it is consumed
Subscribers_Mn	Forecasted level + Delta	EOQ: annual D | WW & SM: demand vector d_1 ... d_T
MoM_Growth_%	Demand volatility sigma	Safety stock for EOQ; validates use of WW/SM over EOQ when sigma is
Market_Share_%	Allocation key	Splits sector-level procurement across the 6 operators
Year, Month	Period index	Defines t = 1...T for the WW DP and SM scan
51 IQR outliers	Demand shocks	Inflate z (EOQ) or kept as spikes in d (WW, SM)
Per-op mean/std	Routing decision	Mature, smooth -> EOQ. Growth-phase -> WW. Lumpy small -> SM

1. Economic Order Quantity (EOQ)

Formula

$$Q^* = \sqrt{2 \cdot D \cdot S / H} \qquad ROP = d_L + z \cdot \sigma_L$$

Inputs: D (annual demand), S (cost per order), H (holding cost per unit per year), sigma (demand std-dev), L (lead time), z (service-level z-score). Two of these come from our dataset (D, sigma); S, H, L come from procurement contracts in P3.

Feature -> EOQ input

Our feature	Role	How it is consumed
Subscribers_Mn (level)	Forecasted D	Forecast next-year monthly subs, take Delta vs previous month -> SIMs/m
MoM_Growth_%	Demand variability sigma	Std of MoM net additions becomes sigma for the safety-stock term
51 IQR outliers	Service-level buffer	Lift z or inflate sigma during shock periods
Market_Share_%	Allocation key	Splits total sector procurement across the 6 operator EOQ runs
Year, Month	Validity check	EOQ assumes constant demand; use trend slope to verify
Per-operator stats	Routing	Picks EOQ for mature low-volatility operators (GP, BL, Robi post-2013)

Worked example - Grameenphone SIMs (2008-2015)

From the stats slide: Grameenphone went 16.88 -> 56.54 M subscribers across 95 month-steps.

```
Step 1 - Demand D from Subscribers_Mn
  Avg monthly net adds = (56.54 - 16.88) / 95 = 0.4175 M  ~= 417,500 SIMs/month
  Annual demand D = 417,500 x 12 ~= 5,010,000 SIMs / year

Step 2 - sigma from MoM_Growth_%
  Std of monthly net additions ~= 350,000 SIMs / month

Step 3 - Cost assumptions (procurement contract, P3)
  S = $500 per order      H = $0.10 / SIM / year
  L = 1 month             z = 1.65 (95% service level)

Step 4 - Plug into the formula
  Q* = sqrt( 2 * 5,010,000 * 500 / 0.10 ) = sqrt( 50,100,000,000 ) ~= 223,830 SIMs
  Orders/year  = 5,010,000 / 223,830 ~= 22  (every ~2.4 weeks)
  Cycle stock  = Q*/2 ~= 111,900 SIMs
  Lead-time demand d_L = 417,500
  Safety stock  = z * sigma = 1.65 * 350,000 ~= 577,500
  Reorder point ROP = 417,500 + 577,500 ~= 995,000 SIMs

Step 5 - Use Market_Share_% to allocate the sector buy
  GP's share of the central order = 43.3%; rerun EOQ per operator and sum.
```

Where each feature literally shows up

```
Subscribers_Mn (forecast) ----------+
Delta Subscribers per month  -------+--> D ---+
                                               |     Q* = sqrt(2 D S / H)
MoM_Growth_%   -->  sigma  -->  SS=z*sigma -+
                                               |     ROP = d_L + SS
IQR outliers   -->  inflate z or sigma -----+
                                               |
Market_Share_% -->  allocation across ops --+
Mean/Std stats -->  pick EOQ vs WW/SM ------+
```

When EOQ is NOT valid

- Growth-phase years (2008-2012) - demand ramps 3x, EOQ overshoots early and undershoots late.
- Small lumpy operators (Citycell, Teletalk) - sometimes negative MoM.
- Outlier-heavy months - constant-demand assumption breaks down.

2. Wagner-Whitin (Dynamic Programming)

Recursion

$$F(t) = \min_{1 \le j \le t} \{ F(j-1) + S + h \cdot \Sigma_{i=j..t} (i - j) \cdot d_i \}, \quad F(0) = 0$$

Reads as: the cheapest plan covering periods 1..t is the cheapest plan ending at j-1, plus **one order in period j** that covers j through t. A DP sweep returns the **globally optimal** order schedule. Inputs: demand vector $d_1..d_T$, setup cost S, per-period holding cost h.

Feature -> WW input

Our feature	Role	How it is consumed
Subscribers_Mn + Delta	Demand vector d	One value per month -> 96 entries (or forecast horizon) per operator
Year, Month	Period index	Defines t = 1..T
MoM_Growth_%	Validates use of WW	Non-stationary growth means EOQ is wrong -> WW handles it
51 IQR outliers	Kept as spikes in d, not smoothed	WW handles shocks period-by-period; no averaging needed
Market_Share_%	Allocation key	Sector demand -> operator-level d vector
Per-operator stats	Routing	Best for growth-phase years 2008-2012 (all operators)

Worked example - first 5 months of Grameenphone 2008

Subscriber readings from the dataset (Jan -> Jun 2008): 16.88, 17.20, 17.81, 18.60, 19.58, 20.31 (millions). Monthly net additions in **thousands of SIMs**:

```
d = [ 320,   610,   790,   980,   730 ]    <- Feb, Mar, Apr, May, Jun
S = $500/order      h = $0.50 / SIM / month
```

Per-segment cost $c(j,t) = S + h \cdot \Sigma(i-j) \cdot d_i$:

order in -> covers	cost ($)
P1 -> 1 only	500
P1 -> 1-2	805
P2 -> 2 only	500
P3 -> 3 only	500
P3 -> 3-4	990
P4 -> 4-5	865
P5 -> 5 only	500

DP sweep:

```
F(0) = 0
F(1) = 500
F(2) = min(805,  500+500)                = 805     (order P1, covers 1-2)
F(3) = min(1595, 500+895, 805+500)       = 1305    (...   + order P3)
F(4) = min(3065, 2375, 1795, 1805)       = 1795    (...   + cover 3-4)
F(5) = min(4525, 3500, 2525, 2170, 2295) = 2170    (P1: 1-2, P3, P4: 4-5)
```

WW optimal plan — 3 orders:

- Feb: order 930K SIMs (covers Feb + Mar)
- Apr: order 790K SIMs (covers Apr)
- May: order 1,710K SIMs (covers May + Jun)
- **Total cost = $2,170** (provably optimal)

Where each feature literally shows up

```
Delta Subscribers_Mn per month --> d_1, d_2, ..., d_T --> DP recursion F(t)
Year / Month                   --> period index t --------+
MoM_Growth_% volatility        --> why WW (not EOQ)       |
51 outlier months              --> kept as spikes in d    |
Market_Share_%                 --> per-operator d vector  --> optimal schedule
```

When WW is the right pick

- Growth-phase years for all operators (2008-2012) - steep, non-stationary trend.
- Forecast horizons where ARIMA / LSTM / XGBoost output a clearly time-varying vector.
- Computationally fine for T up to thousands (O(T^2)); 96 months is trivial.
- Optimal but deterministic - combine with safety stock from MoM_Growth_% sigma for uncertainty.

3. Silver-Meal Heuristic

Rule

$$TRUC(k) = (S + h \cdot \Sigma_{i=1..k} (i-1) \cdot d_{j+i-1}) / k$$

TRUC = Total Relevant Unit Cost *per period covered*. Keep increasing k while TRUC keeps falling. The moment TRUC(k+1) > TRUC(k), **stop** - the current order covers periods j..j+k-1. Then restart at j+k. Greedy, no DP.

Feature -> SM input

Our feature	Role	How it is consumed
Delta Subscribers per month	Demand vector d	Same as WW, but SM only looks forward, never reconsiders past breakpo
MoM_Growth_% low / lumpy	Best fit signal	SM shines for low-volume + irregular demand (Teletalk, Citycell)
51 IQR outliers	Trigger early stop	A shock pushes TRUC up -> SM closes the lot immediately, surge-aware
Market_Share_%	Allocation key	Same per-operator d vector as WW
Year, Month	Forward scan index	Drives the look-ahead loop
Per-operator stats	Routing	Best for small lumpy operators (Teletalk, Citycell) and rolling re-plans

Worked example - same GP 2008 vector

```
d = [ 320, 610, 790, 980, 730 ]    S = $500     h = $0.50
```

Starting at P1 (Feb):

```
TRUC(1) = 500 / 1                            = 500.00
TRUC(2) = (500 + 0.5*(0*320 + 1*610)) / 2    = 402.50    DOWN  continue
TRUC(3) = (500 + 0.5*(610 + 2*790)) / 3      = 531.67    UP    STOP
```

Place order in **Feb covering Feb + Mar** -> Q = 930K SIMs. Restart at P3.

Starting at P3 (Apr):

```
TRUC(1) = 500
TRUC(2) = (500 + 0.5*(0 + 1*980)) / 2        = 495.00    DOWN  continue
TRUC(3) = (500 + 0.5*(980 + 2*730)) / 3      = 573.33    UP    STOP
```

Place order in **Apr covering Apr + May** -> Q = 1,770K SIMs. Restart at P5.

Starting at P5 (Jun): only one period left -> single order of 730K SIMs, cost $500.

SM plan total cost: $805 + $990 + $500 = **$2,295**

Where each feature literally shows up

```
Delta Subscribers_Mn per month --> d_j, d_{j+1}, ...
                                        |
                                        v   (look forward k periods)
                                   compute TRUC(k)
                                        |
                                        v   (stop when TRUC rises)
MoM_Growth_% volatility            -->  short, frequent lots
51 outliers                        -->  early stop, natural surge handling
Market_Share_%                     -->  per-operator d vector
```

When SM is the right pick

- Low-volume, lumpy operators (Teletalk, Citycell) where running 96-period DP is wasteful.
- Rolling-horizon use: SM only needs the next few months of forecast - good for online re-planning.
- Acceptable cost gap: SM lands within ~1-7% of Wagner-Whitin on most demand shapes.

Side-by-side on the same 5-month GP slice

	Setup cost	Holding cost	Total cost	# orders
EOQ (avg only)	varies	varies	n/a	~22/yr
Wagner-Whitin	$1,500	$670	$2,170	3
Silver-Meal	$1,500	$795	$2,295	3

Wagner-Whitin wins by ~5.7% but pays for it in computation. Silver-Meal is fast and *good enough*. EOQ is shown for reference - it should not be used on this growth-phase slice because demand is clearly non-stationary.

How ChronoForge Pulse routes operators to algorithms

Operator profile (from EDA)	Algorithm	Reason
Mature, smooth, low MoM volatility (GP / BL / Robi post-2013)	EOQ	Constant-demand assumption holds; analytic formula is fastest
Growth-phase, steep non-stationary trend (all operators 2008-2012)	Wagner-Whitin	Optimal for time-varying d; pays off where the trend is steep
Small, lumpy, sometimes-negative MoM (Teletalk, Citycell)	Silver-Meal	DP is wasteful here; heuristic is within ~5% of optimum

Bottom line: Every lot-sizing input comes back to two engineered features — **Subscribers_Mn (level + delta)** and **MoM_Growth_%** — plus the allocator **Market_Share_%**. That is exactly why Presentation 2 invests so heavily in cleaning, tidying, and engineering these features: they are the literal numerical inputs to every one of the three algorithms shipped with ChronoForge Pulse.